# Practical-8

# MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

## Theory

### 1. Is the schema normalized properly?

**Answer:** Yes. The database is divided into separate tables like students, events, and registrations. This avoids repeating the same data and keeps the database organized.

### 2. Are primary keys and foreign keys correctly defined?

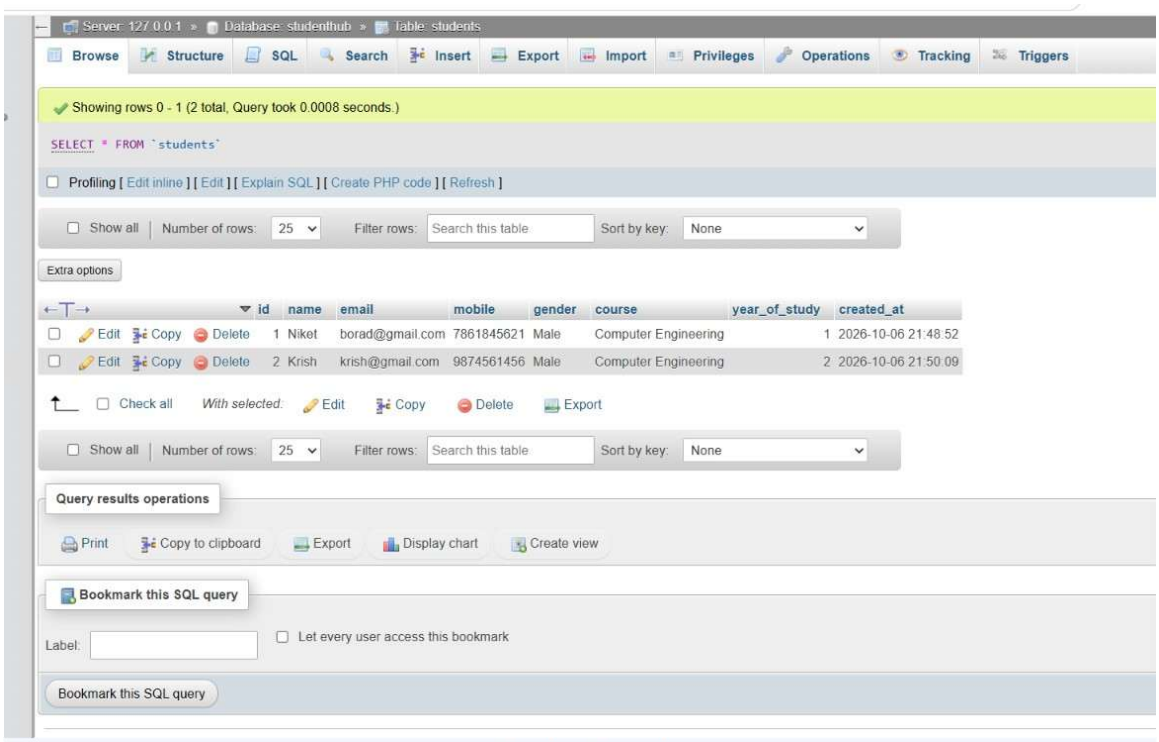
**Answer:** Yes. Each table has a **primary key** to uniquely identify each record. Foreign keys connect related tables, such as student_id and event_id in the registrations table.

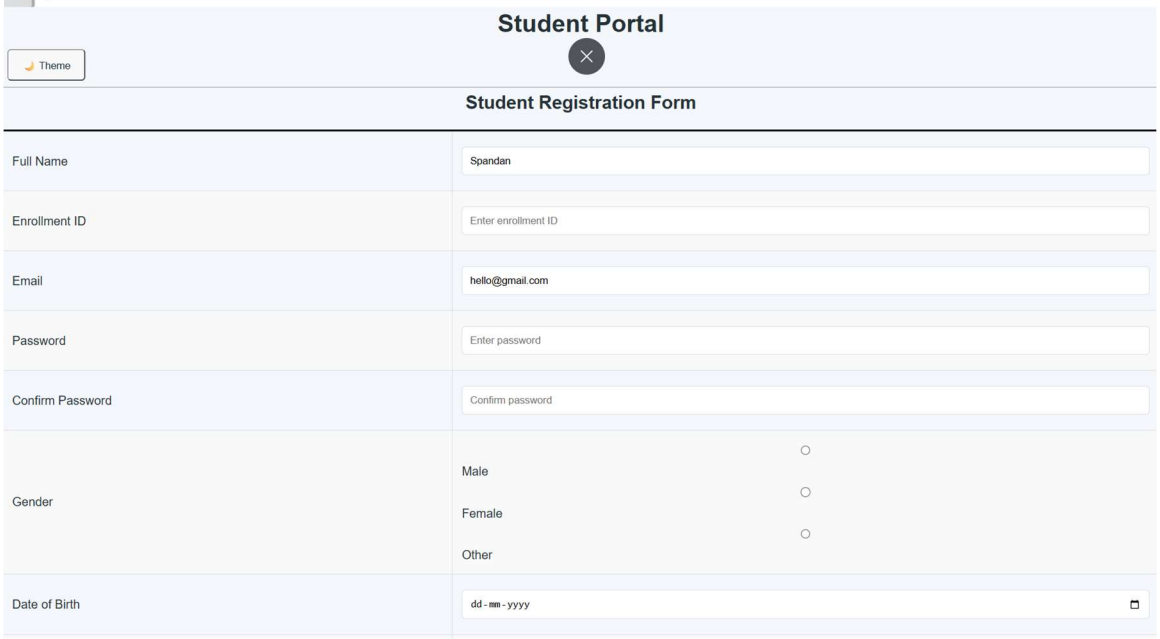
### 3. Is the database connection handled securely?

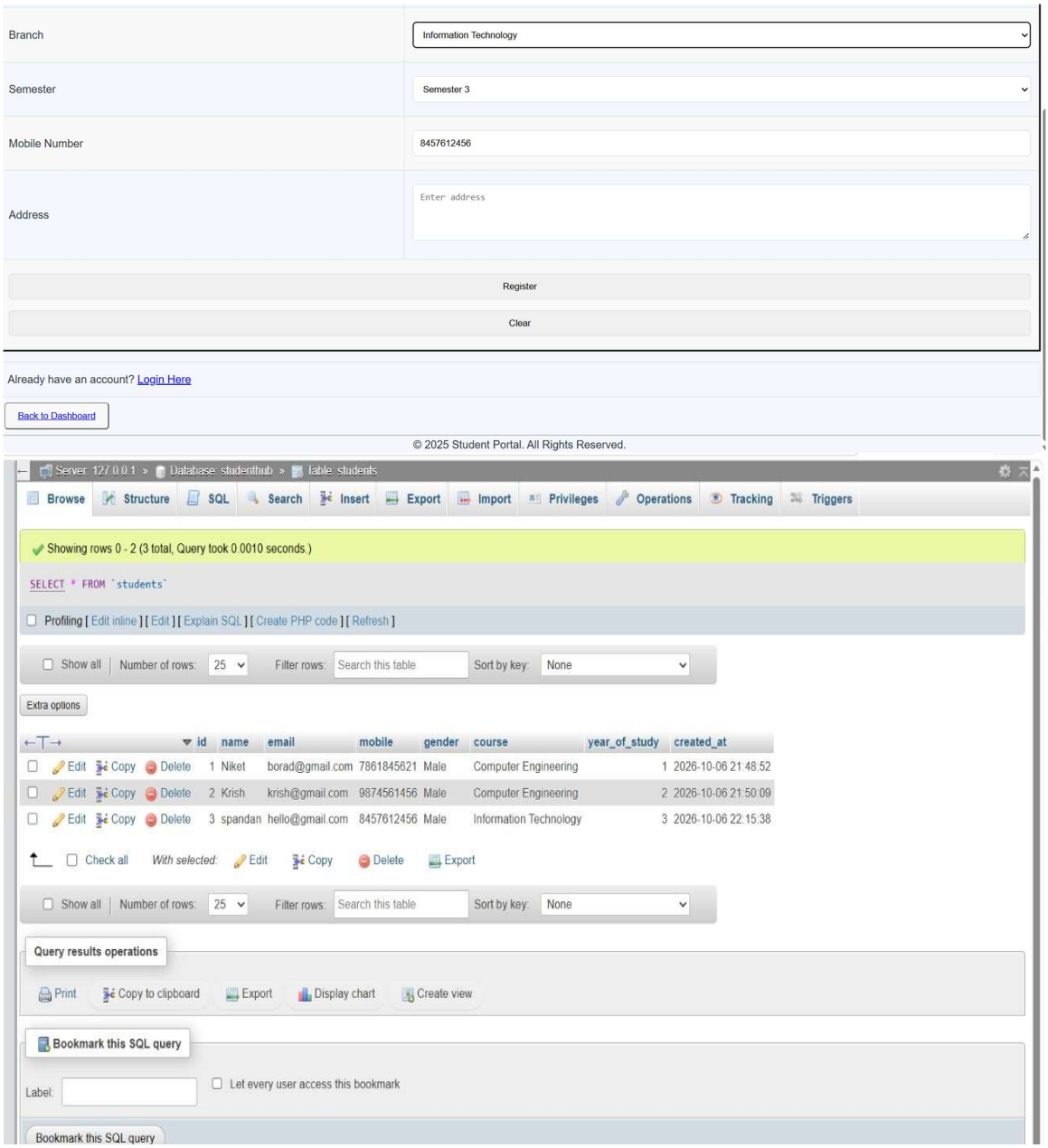
**Answer:** Yes, reasonably. The connection details are kept in a separate db.php file, and the code uses prepared statements when inserting data.

## Result:

### Evidence of Work (Screenshots)


Server: 127.0.0.1 » Database: studenthub » Table: students
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Tracking
Triggers
Showing rows 0 - 1 (2 total, Query took 0.0008 seconds.)
SELECT * FROM `students`
Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id name email mobile gender course year_of_study created_at
Edit Copy Delete 1 Niket borad@gmail.com 7861845621 Male Computer Engineering 1 2026-10-06 21:48:52
Edit Copy Delete 2 Krish krish@gmail.com 9874561456 Male Computer Engineering 2 2026-10-06 21:50:09
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Bookmark this SQL query


Student Portal
Theme
Student Registration Form
Full Name
Spandan
Enrollment ID
Enter enrollment ID
Email
hello@gmail.com
Password
Enter password
Confirm Password
Confirm password
Gender
Male
Female
Other
Date of Birth
dd-mm-yyyy




## A. GitHub Repository Link:

https://github.com/krish-301/Student-Hub-Portal

For XAMMP Server Link:-

http://localhost/Student%20port/Pages/registration.html

## B. Challenges Faced:

I did not know much about MYSQL at the beginning.
I had difficulty understanding PHP syntax. And How to connect with MYSQL.
I was confused about how to connect my HTML pages with MYSQL.
I did not know how to run my project using the XAMPP server. Also How to Make Table in XAMMP Server also diificult for beginner .
I faced errors while connecting the HTML form with the PHP file. And MYSQL.

## C.Learning Outcome:-

Learned to design a normalized MySQL database, connect PHP with MySQL, and use primary/foreign keys and prepared statements securely.

Faculty Remarks: